

Long-context DNA models do not use the allele: an audit of DNALongBench’s eQTL task

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Abstract

eQTL prediction (eQTLP) is the only variant task in DNALongBench (Cheng et al., 2025), a standard benchmark for long-context DNA models. Ranking variants by distance to the transcription start site (TSS), with no training, reaches a mean test AUROC of 0.751, beating the expert model (0.681) and every foundation model (FM; 0.51–0.57). The test sets hold 11–34 positives per tissue, and every test gene also appears in training. We rebuild the task with distance-matched negatives and chromosome holdout, where hand-crafted local sequence features set a floor of 0.599. Scoring each pair again with the alternate allele replaced by the reference shows that no model uses the allele: for a CNN trained end to end and for frozen HyenaDNA and Caduceus, the swap changes AUROC by at most 0.0017, and all three fall below the floor. Fine-tuning HyenaDNA end to end over 3 folds does not change this: accuracy improves to the frozen probe’s level while the allele gap stays at $|\Delta| \leq 0.00016$. A nonlinear probe on the allele difference alone recovers only a weak trace (0.52–0.55). A single-nucleotide change moves the pooled FM embedding by about 10^{-4} of its norm, and the released FM eQTL code slices every input to length zero. On 652 ClinVar pathogenic regulatory SNVs matched to benign SNVs, neither the embedding shift nor the transferred probe separates the classes (AUROC 0.49–0.51). We release the rebuilt task, the ablation harness, and the matched ClinVar set.

1 Introduction

Long-context DNA models (Nguyen et al., 2023; Schiff et al., 2024; Brixi et al., 2026) are motivated by distal regulation, meaning variants that act on genes tens to hundreds of kilobases away. They inherit their architectures from long-convolution and state-space sequence models (Poli et al., 2023; Gu and Dao, 2023), and are benchmarked against expert models such as Enformer (Avsec et al., 2021), Akita (Fudenberg et al., 2020) and Puffin (Dudnyk et al., 2024). DNALongBench (Cheng et al., 2025) evaluates HyenaDNA (Nguyen et al., 2023) and Caduceus (Schiff et al., 2024) at up to 450 kb. Of its five tasks, only eQTLP involves a variant. Its model interface embeds the reference and alternate sequences, averages each, concatenates them, and classifies. A model can therefore succeed without distinguishing the alleles at all, if *where* the variant sits predicts the label.

We show that this is what happens. The published split rewards distance and gene identity. Once both are removed, three architectures score the same with and without the alternate allele. We contribute: (1) an audit of the published protocol; (2) **eQTLP-v2**, a distance-matched, chromosome-held-out version with a published floor; (3) a **ref-copy ablation** that any variant benchmark can report; (4) reproducibility findings on the released code; and (5) a matched, evaluation-only ClinVar regulatory set.

2 The published protocol rewards distance and memorization

Positives are SuSiE (Wang et al., 2020) fine-mapped GTEx (The GTEx Consortium, 2020) eQTLs in 9 tissues. Negatives are other variant–gene pairs within 450 kb. Each input spans the variant and the TSS and is right-padded with N to 450 kb. The split is a per-tissue random 8:1:1.

- **Distance beats every model.** Positives lie a median 8–33 kb from the TSS, negatives 88–145 kb. Scoring by $|\text{variant} - \text{TSS}|$ from the released coordinates, which is what the padded input encodes) gives a test AUROC of 0.688–0.828 per tissue, with a

Table 1: Published test AUROC (Table 7 of the benchmark paper), mean over 9 tissues, next to our zero-training baseline. Per-tissue values with CIs are in Appendix A.

Distance only	Expert (Enformer)	CNN	HyenaDNA	Caduceus-Ph	Caduceus-PS
0.751	0.681	0.528	0.514	0.566	0.538

mean of 0.751 (Table 1). That beats the expert model in 7 of 9 tissues and every FM in every tissue. The padding also hands the distance to the model, since the number of N tokens encodes it.

- **The test sets cannot rank models.** They hold 11–34 positives per tissue, giving AUROC CI half-widths of ± 0.10 – 0.18 (Hanley and McNeil, 1982), so no published FM result is distinguishable from 0.5. Nor is the split stratified: positive rates are 9–21% in train, 3–5% in test.
- **Train and test overlap.** Every test gene appears in training, and 82–92% of test pairs have a training variant within 100 kb.

3 eQTL-v2

We pool the tissues into 25,600 unique pairs and drop 111 with conflicting labels. Each positive is matched to one negative on log distance to the TSS, on the same side of it, which gives 2,905 pairs and drops the distance-only AUROC to 0.495. Folds are 5-fold with whole chromosomes held out, giving 570–665 test positives per fold. We also drop the published label-derived mask, which covers the positions of the positive eQTLs.

Matching on distance removes real regulatory signal along with the shortcut, since enhancer–promoter distance genuinely predicts effect. We match anyway, because the task’s stated aim is variant-effect prediction from sequence, and because the published input encodes distance in its padding, so a model can exploit it without reading any sequence. Unmatched results are in Appendix B.

On this configuration, gradient-boosted trees on local composition (GC, CpG, repeats and dinucleotides in windows up to 5 kb) score **0.599 $\pm$ 0.031**. Adding allele features changes this by at most 0.01. On the random split, a gene’s training-set positive rate alone scores 0.70. That is the memorisation the published split allows.

4 No model uses the allele

Protocol. Each model embeds the reference and alternate sequences. The last-layer states are mean-pooled, and $[e_{\text{ref}}, e_{\text{alt}}]$ is classified, as in the benchmark’s Methods. We report three quantities: **refalt**, the test AUROC; **ref_copy**, the same fitted classifier scoring $[e_{\text{ref}}, e_{\text{ref}}]$; and **diff_only**, a classifier trained only on $e_{\text{alt}} - e_{\text{ref}}$. The models are the benchmark’s **CNN** encoder, trained end to end with its collapsing logit ReLU repaired (§5); and **HyenaDNA-medium-450k** (Nguyen et al., 2023) (causal, 6.55 M parameters) and **Caduceus-Ph-131k** (Schiff et al., 2024) (bidirectional, 7.73 M), frozen, with a logistic-regression probe (Pedregosa et al., 2011). Inputs are unpadded. For each FM we embed both the published orientation and its reverse complement.

- **The allele changes nothing** (Figure 1a). Replacing it with the reference changes AUROC by at most 0.0017 for the trained CNN (15 runs) and 0.0006 for the FM probes, across probe strengths ($C = 0.001$ – 10), heads (linear, MLP), orientations, and 13 fold partitions (chromosome, gene and random). The paired 95% CI is $[-0.0002, 0.0002]$.
- **Only a weak trace of the allele is there to find, and pooling is not what hides it.** A linear **diff_only** probe scores at chance (0.49–0.51). An MLP on the difference beats a label-permutation null (Caduceus 0.54–0.55, HyenaDNA 0.52–0.53) but stays below the floor, and an MLP head on $[e_{\text{ref}}, e_{\text{alt}}]$, like the released Caduceus head, still has $\Delta = 0.0000$.

Averaging over the whole sequence discards position, so we next ask whether the allele

Table 2: eQTLP-v2, matched pairs, chromosome holdout. Mean per-fold test AUROC $\pm$ SD.

Model	refalt	ref_copy	Δ	diff_only
CNN (trained, 3 seeds $\times$ 5 folds)	0.517 $\pm$.040	0.517	-0.0002	—
HyenaDNA (frozen)	0.556 $\pm$.031	0.556 $\pm$.030	-0.0001	0.495 $\pm$.021
Caduceus-Ph (frozen)	0.564 $\pm$.021	0.564 $\pm$.021	0.0000	0.507 $\pm$.016
Local-composition floor	0.599 $\pm$.031			

survives somewhere that average cannot reach. We repeat the search at every layer and over four *summary windows* — the set of positions averaged together before classifying: the whole sequence, ± 1 kb around the variant, ± 64 bp, and the single variant position itself. Each is scored against a control in which the variant is replaced by a base that did **not** occur, which absorbs anything attributable to local sequence context rather than to the allele. For HyenaDNA the control matches the signal everywhere (mean -0.006 over 36 cells), so what looked like allele information was context. For Caduceus one window survives its control — the final layer at ± 1 kb, scoring 0.575 – 0.580 against 0.545 – 0.553 , on all three fold schemes. So a variant-centred summary does recover allele signal that whole-sequence averaging loses. It is still below the floor, and the swap still changes nothing even there (max $|\Delta|$ 0.017 over 104 cells, always favouring **ref_copy**).

- **All three models score below the hand-feature floor** (Figure 1a), in 10 of 10 random chromosome partitions. Only on the random split, which allows locus memorisation, do the FMs pass it (0.69 – 0.70 vs 0.67).

The SNP barely registers (Figure 1b). The median relative shift $\|e_{\text{alt}} - e_{\text{ref}}\|/\|e_{\text{ref}}\|$ is 1.5 – 2.0×10^{-4} for HyenaDNA and 0.7×10^{-4} for Caduceus. eQTLP-v2 is 92.3% single-nucleotide substitutions and 7.7% short indels or MNVs; restricting to the substitutions alone leaves both medians unchanged (1.52×10^{-4} and 0.674×10^{-4}), so this is a per-variant effect and not an artifact of the indels. The CNN’s max-pooled embedding changes in only 0.4–1.7% of pairs. Two explanations for this fail.

Orientation: HyenaDNA is causal, and in the published orientation most variants sit in the last 500 bp; but reversing the input leaves AUROC unchanged, and bidirectional Caduceus behaves identically. **Precision:** re-embedding 400 pairs in fp32 gives the same shift (bf16–fp32 difference-vector cosine 0.89 – 0.95), and **diff_only** stays at chance. What remains is pooling. The SNP moves the variant-position hidden state by 37–112%, the ± 64 bp mean by 0.7–6.1%, and the whole-sequence mean by 0.006–0.09%: mean-pooling over tens of kilobases dilutes the substitution $\sim 1,000\times$.

Fine-tuning does not change it. The probes above are frozen, whereas the published numbers come from fine-tuning, so we fine-tuned HyenaDNA-medium-450k end to end on eQTLP-v2 — backbone and head, through the benchmark’s own mean-pooled readout — for 2 epochs on each of 3 chromosome folds (33 GPU-hours; inputs capped at 131 kb, which 88% of pairs fit entirely).

Training works — loss falls $0.730 \rightarrow 0.698$ and **refalt** rises on 2 of 3 folds — and lands at 0.555 ± 0.023 , indistinguishable from the frozen probe’s 0.556 ± 0.031 and still below the floor. The allele gap does not move: across all six epochs the largest $|\Delta|$ is **0.00016**. The model improves at the task while learning nothing about which allele is present, so the null is not an artifact of freezing the backbone.

Scope and limitations. The fine-tuning run is one model, one seed, 2 epochs per fold at 131 kb rather than the published 450 kb, so it bounds what this much training achieves, not what any training could. We evaluate two models and both are small (6.55 M and 7.73 M parameters); larger long-context models such as Evo 2 (Brix et al., 2026) are beyond our compute. Our claim is therefore about what these architectures and this readout do on this task, not about model scale.

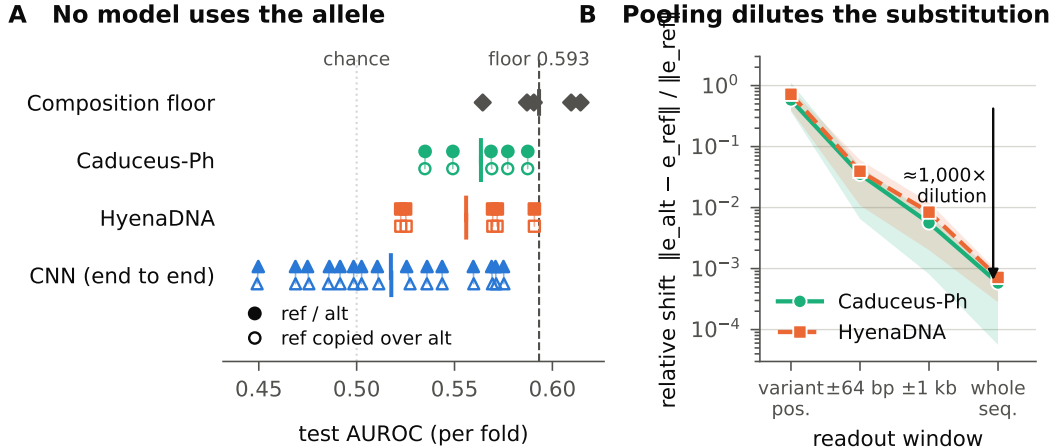


Figure 1: (a) eQTL-v2, matched pairs, chromosome holdout: per-fold test AUROC for each model, scored normally (**refalt**, filled) and with the reference copied over the alternate allele (**ref_copy**, open). Each fold’s pair is joined by a line, and every line is vertical: the swap moves no score. Points are 5 folds ($\times 3$ seeds for the CNN); the bar is the mean. The dashed line is the floor on these same folds (0.593); §3 quotes 0.599 ± 0.031 , its mean over 7 matchings $\times 10$ partitions. No model reaches it. (b) Relative shift $\|e_{alt} - e_{ref}\| / \|e_{ref}\|$ under four summary windows, from the variant position out to a whole-sequence average; markers are the median over layers, bands the min-max.

5 The released code does not run as shipped

We report defects found by reading and running the released code; we do not claim to have re-run the published experiments, since with these defects the released code cannot produce the published FM numbers at all.

- **The FM eQTL loaders produce empty inputs.** Both the HyenaDNA and Caduceus loaders pad each input to 450,000 positions, then slice it with `[2028500:-2028500]`, leaving an empty array.
- **The token ids are wrong.** Without the slice, `argmax` over the one-hot gives ids 0–3, which are special tokens in the pretrained vocabularies.
- **The configs don’t match the checkpoints.** They specify 2 layers, while the checkpoints have 8 and 16, and the pretrained-weights path is unset.
- **The CNN’s training often collapses.** Its eQTL head ends in a ReLU on the logits, so both logits can clamp to zero: the prediction is constant and every weight gets zero gradient. On the *published* split, 2 of 10 initialisations are dead before the first step, and after one epoch 5–6 of 10 seeds output a constant score (AUROC exactly 0.5). The script sets no seed and evaluates the final-epoch model, not the best. The CNN is fed only the reference.

6 Clinical evaluation set

ClinVar (Landrum et al., 2025) pathogenic or likely-pathogenic regulatory SNVs with at least 1 star are filtered to non-coding variants over 50 bp from a splice site, off every CDS and outside small structural RNAs; 652 map to a gene TSS within 450 kb, each matched to a benign SNV passing the same filters on distance to the TSS and, where possible, gene and location class. No model is fitted on clinical labels, and the embedding shift, the transferred eQTL-v2 probe and its allele-only component are all at chance (0.489–0.507). The only stratum above chance is 5’UTR/promoter (0.53–0.57), where matching left pathogenic variants closer to the TSS and distance alone scores 0.61. The same null therefore appears under pathogenicity labels; the two targets differ, so this corroborates the eQTL result rather

than generalising it.

7 Relation to prior work

Two long-range DNA benchmarks are adjacent to ours: the Genomics Long-Range Benchmark (Kao et al., 2024), with an OMIM pathogenic non-coding variant task, and BEND (Marin et al., 2024), on reference-sequence element annotation. Neither reports an allele ablation, the diagnostic we argue for: without it, a variant task can be passed by locating the variant rather than reading it. Our result also extends to a benchmark setting the finding that sequence-to-expression models predict individual expression variation poorly (Sasse et al., 2023; Huang et al., 2023). The failure mode is shortcut learning (Geirhos et al., 2020): the protocol, not the architecture, decides what a model must learn.

8 Recommendations

Variant benchmarks for long-context models should match negatives on distance to the TSS and, where possible, gene; hold out chromosomes; size test sets for ± 0.05 CIs; publish a hand-feature floor; report a **ref-copy ablation** beside every variant score; and read the variant out *locally* rather than as a whole-input mean.

On DNALongBench’s eQTL task, current models learn where a variant is, not what it is.

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Appendix

A per-tissue Table 1 with CIs, test-set sizes and overlap statistics; **B** eQTL-v2 construction and the full baseline grid (random, gene and chromosome holdout; all vs matched); **C** per-fold ablation results, results by orientation, and the fp32 check; **D** the CNN collapse analysis; **E** the released-code defects with file and line references; **F** clinical-set construction and results by stratum.